Problem 12 | Problems T5_MechanicalProperties.pdf #12 | True stress · true strain

# Steel Cable: True Fracture Stress and True Strain from the Reduction in Area

Original statement · Problems T5_MechanicalProperties.pdf, p. 5

*12. A steel cable has fracture strength of 300 MPa. Its ductility is 77% measured as a reduction in area. Calculate the real fracture strength. Considering that strain is uniform until fracture, what is the real strain?*

**Official Solution:** $\varepsilon_{real} = 1.47$

PHASE 1 · HYPOTHESES & PARAMETERS

### Hypotheses

1. **Fracture strength = engineering stress at fracture.** The "fracture strength of 300 MPa" is the load at fracture divided by the *initial* area, $\sigma_f = F_f / A_0$, as in the engineering curve of the tensile test [Session 11 Slide 9]. The true fracture stress is that load divided by the area at fracture, $A_f$ [Session 11 Slide 37].
2. **Ductility as reduction in area.** $RA\,(\%) = \dfrac{A_0 - A_f}{A_0} \times 100$ is measured after fracture [Session 11 Slide 33], so $RA = 77\,\%$ means $A_f = 0.23\,A_0$.
3. **Uniform deformation.** The statement asks to take the strain as uniform until fracture, so the volume of the wire is constant, $l_0 A_0 = l_f A_f$ (the necking is not modelled) [Session 11 Slide 38].
4. **Real strain by integration.** $\varepsilon_r = \int_{l_0}^{l} dl/l = \ln(l/l_0)$ [Session 11 Slide 37].

### Data

| Quantity | Symbol | Value | In SI |
|---|---|---|---|
| Engineering fracture strength | $\sigma_f$ | 300 MPa | $300 \times 10^6$ Pa |
| Reduction in area | $RA$ | 77 % | 0.77 (dimensionless) |

**Unknowns:** the true fracture stress $\sigma_{f,r}$ and the true strain at fracture $\varepsilon_r$.

PHASE 2 · FRAMES & STRATEGY

One direction (the axis of the wire); no change of basis.

**Strategy.**

1. The reduction in area gives the ratio $A_f/A_0 = 1 - RA$.
2. The load at fracture is the same in both definitions of stress, $F_f = \sigma_f A_0 = \sigma_{f,r} A_f$, so $\sigma_{f,r} = \sigma_f \, A_0/A_f$ (the true stress is larger than the engineering one because the area has decreased) [Session 11 Slide 38].
3. With constant volume, the length ratio is the inverse of the area ratio, and the true strain follows from $\ln(l/l_0) = \ln(A_0/A_f)$.
4. The result is verified with the other route, $\sigma_r = \sigma(1+\varepsilon)$ and $\varepsilon_r = \ln(1+\varepsilon)$, through the engineering strain.

PHASE 3 · STEP-BY-STEP DERIVATION

**1. Area ratio at fracture**

$$\frac{A_f}{A_0} = 1 - \frac{RA}{100} = 1 - 0.77 = 0.23$$

**2. True fracture stress**

The load at fracture is $F_f = \sigma_f A_0$. Dividing by the area at fracture instead of the initial one:

$$\sigma_{f,r} = \frac{F_f}{A_f} = \frac{\sigma_f A_0}{A_f} = \frac{\sigma_f}{A_f/A_0} = \frac{300\ \text{MPa}}{0.23} = \mathbf{1304.3\ MPa}$$

($\sigma_{f,r}/\sigma_f = 4.348$: the true stress is more than four times the engineering one.)

**3. True strain at fracture**

With the volume constant, $l_0 A_0 = l_f A_f$, so

$$\frac{l_f}{l_0} = \frac{A_0}{A_f} = \frac{1}{0.23} = 4.348$$

and the integral of the definition of the real strain, with the substitution $u = l$, $du = dl$, and Barrow's rule between $l_0$ and $l_f$, is

$$\varepsilon_r = \int_{l_0}^{l_f} \frac{dl}{l} = \left[\ln l\right]_{l_0}^{l_f} = \ln l_f - \ln l_0 = \ln \frac{l_f}{l_0} = \ln \frac{A_0}{A_f} = \ln \frac{1}{0.23} = \mathbf{1.4697}$$

Equivalently, in terms of the diameters of the wire, $\varepsilon_r = 2\ln(D_0/D_f)$ [Session 11 Slide 38], with $D_f/D_0 = \sqrt{A_f/A_0} = \sqrt{0.23} = 0.4796$: $2\ln(1/0.4796) = 1.4697$.

**4. Check through the engineering strain**

The engineering strain at fracture is $\varepsilon = l_f/l_0 - 1 = 4.348 - 1 = 3.348$ (that is, 334.8 %). Then

$$\sigma_{f,r} = \sigma_f\,(1+\varepsilon) = 300 \times 4.348 = 1304.3\ \text{MPa} \qquad \varepsilon_r = \ln(1+\varepsilon) = \ln 4.348 = 1.4697$$

in agreement with the two results above.

PHASE 4 · CHECKS & INTERPRETATION

**Checks**

- **Units.** Stress in MPa; $A_f/A_0$ and the strains are dimensionless; the argument of the logarithm is a ratio of areas.
- **Limits.** If $RA \to 0$ (no necking) then $A_f \to A_0$, $\sigma_{f,r} \to \sigma_f$ and $\varepsilon_r \to 0$. As $RA \to 100\,\%$, both $\sigma_{f,r}$ and $\varepsilon_r$ grow without bound.
- **Ordering.** $\sigma_{f,r} > \sigma_f$ and $\varepsilon_r < \varepsilon$ ($1.4697 < 3.348$), as stated in [Session 11 Slide 37].

**Interpretation**

The steel cable breaks when the material that is left carries 1304 MPa, not 300 MPa: the engineering value hides the reduction of the section. The true strain, 1.47, is the natural measure of such a large deformation, because the integral adds up the increments $dl/l$ over the changing length.

LIMITATION OF THE UNIFORM-STRAIN HYPOTHESIS

A real cable necks before fracture, and most of the strain is concentrated in the neck. The mean strain along the whole length, obtained here, is therefore a lower bound of the local strain at the fracture section. The statement asks explicitly for the uniform strain, so that is the value reported.

**Check against the official solution**

| Item | Official | Computed | Comment |
|---|---|---|---|
| true fracture stress | not given | 1304.3 MPa | asked by the statement, absent from the key |
| true strain | 1.47 | 1.4697 | agrees |

**DISCREPANCY WITH THE OFFICIAL SOLUTION: THE KEY OMITS THE TRUE STRESS**

The key only prints "$\varepsilon_{real} = 1.47$", although the statement asks for the real fracture strength first. The value computed here, $\sigma_{f,r} = 1304.3$ MPa, is not contradicted by the key: it is the first half of the answer that the key leaves out.